higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

CONTROL SYSTEMS N6

3 DECEMBER 2019

This marking guideline consists of 8 pages.

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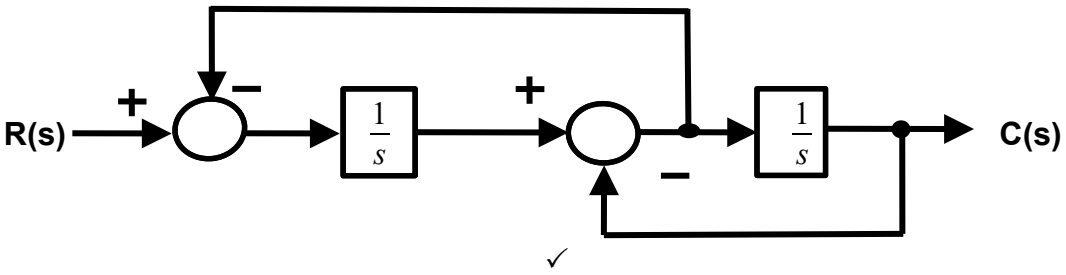
QUESTION 1

1.1 True
1.2 True
1.3 False
1.4 True
1.5 True
1.6 False
1.7 True
1.8 False
1.9 True
1.10 False

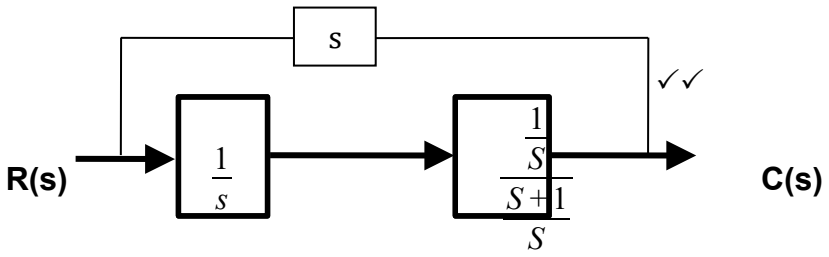
(10 × 1) **[10]**

QUESTION 2

2.1



✓



✓✓

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{s(s+1)}}{1+\frac{1}{s(s+1)}.\frac{s}{1}} \checkmark$$

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{s(s+1)}}{\frac{(s+1)+1}{(s+1)}} \checkmark$$

$$\frac{C(s)}{R(s)} = \frac{\frac{1}{s(s+1)}}{\frac{(s+2)}{(s+1)}} \checkmark$$

(7)

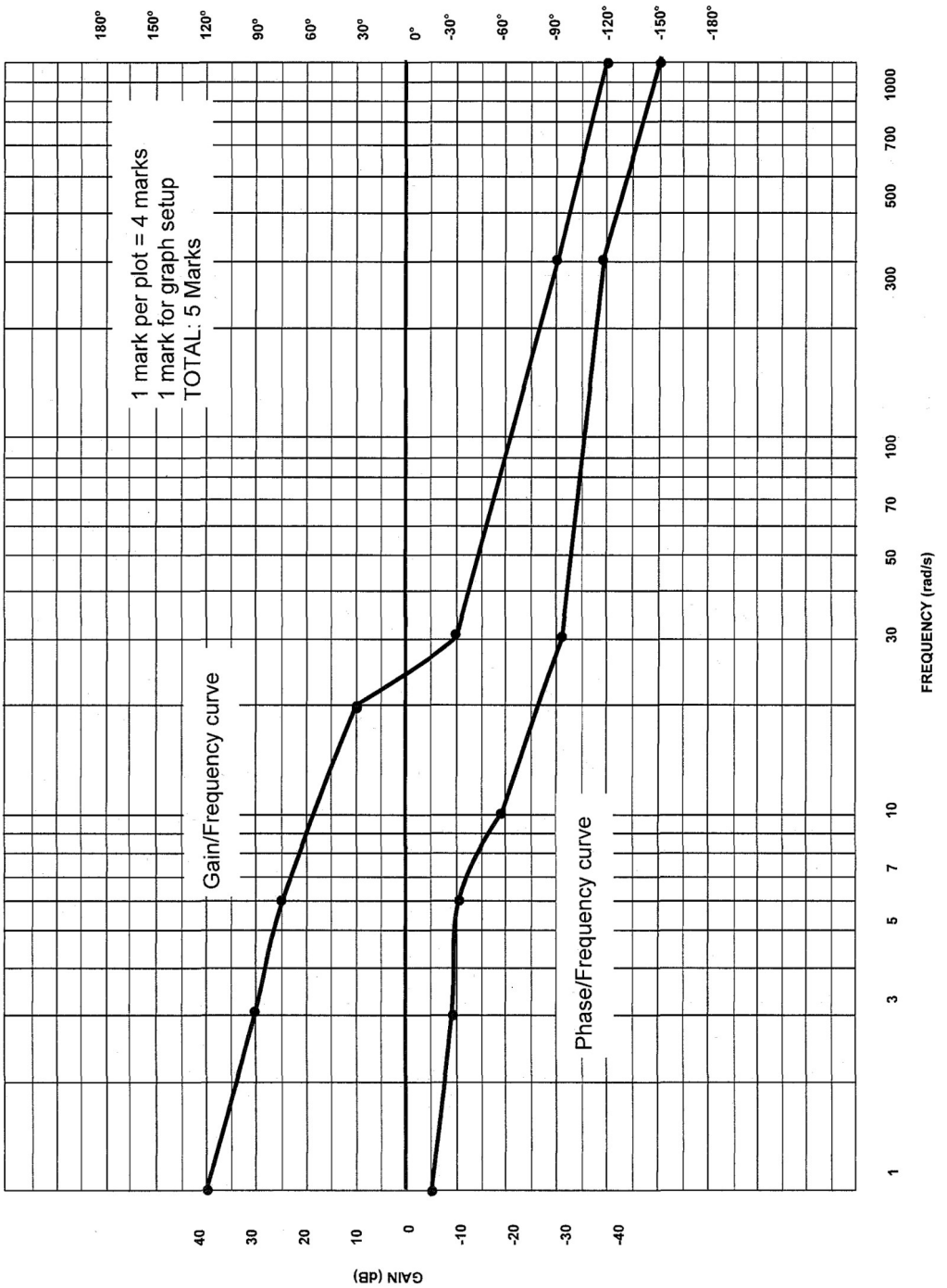
$$\frac{C(s)}{R(s)} = \frac{1}{s} \cdot \frac{1}{(s+2)} \checkmark$$

2.2
- It enters into the process and upsets the system.
- It provides an output different from the set point making the system take corrective action.
- It can seriously affect or damage the control action and create a time delay in the control action if it is of a large magnitude. (3)

[10]

QUESTION 3

3.1



(5)

3.2 3.2.1 25 rad/s (1)

3.2.2 No phase crossover frequency is obtainable as the phase plot does not intersect the -180° line. (2)

3.2.3 90° (1)

3.2.4 The gain margin is not determinable as there is no phase crossover frequency value. (1)
[10]

QUESTION 4

4.1

ω (rad/sec)	5	10	150
Magnitude in dB	0,09	0,37	-11,8
Phase in degrees	-2,6°	-6°	-207°
	✓✓	✓✓	✓✓

(6)

4.2 4.2.1 84 rad/s
4.2.2 112,5 rad/s
4.2.3 3,7 dB
4.2.4 -80°

(4 × 1) (4)
[10]

QUESTION 5

5.1 Poles = 0; -5✓ Zeroes = -7✓ (2)

5.2 $\zeta = \cos\ \theta$
$\theta = \cos^{-1}\ 0{,}707$✓
$\theta = 45°$✓

$\omega_d = 2{,}2\ \ rad/s$✓ (3)

5.3 $\omega_n = \Delta\ P_2 = 3{,}2\ \ rad/s$ (1)

5.4 $\Delta\ P_1 = 3{,}5$
$\Delta\ P_2 = 3{,}2$✓
$\Delta\ Z_1 = 5{,}3$

$$K_o = \frac{\Delta\ P_1 \Delta\ P_2}{\Delta\ Z_1}$$

$$K_o = \frac{3{,}5 \times 3{,}2}{5{,}3}$$

(3)

$K_o = 2,113$✓

5.5 6,1 rad/s (1)
[10]

QUESTION 6

6.1 6.1.1 $F(t) = \frac{10}{49}e^{-7t} + 7t - 1$

$F(s) = \frac{10}{s^2(s+7)}$✓✓

6.1.2 $F(t) = 115\left[1 - \frac{(5+t)}{5}\ e^{\frac{-t}{5}}\right]$

$F(s) = \frac{115}{s(5+a)^2}$✓✓

(2 × 2) (4)

6.2 6.2.1 $F(s) = \frac{30\omega}{(s^2 + \omega^2)(3s+1)}$

$F(t) = \frac{30\omega 3}{1 + \omega^2 3^2}\ e^{\frac{-t}{3}} + \frac{30}{\sqrt{1 + \omega^2 3^2}}\sin(\omega t - \Psi)$✓

$F(t) = \frac{90\omega}{1 + 9\omega^2}\ e^{\frac{-t}{3}} + \frac{30}{\sqrt{1 + 9\omega^2}}\sin(\omega t - \Psi)$✓

where

$\Psi = \tan^{-1} 3\omega \quad (0 < \Psi < \pi)$✓

6.2.2 $F(s) = \frac{24}{s^2(s+4)(s+3)}$

$F(t) = \frac{24}{4 \times 3}\left[t - \frac{4+3}{4 \times 3} - \frac{\frac{3}{4}\ e^{-3t} - \frac{4}{3}\ e^{-4t}}{3-4}\right]$✓

$F(t) = \frac{24}{12}\left[t - \frac{7}{12} - \frac{\frac{3}{4}\ e^{-3t} - \frac{4}{3}\ e^{-4t}}{-1}\right]$✓

$F(t) = 2\left[t - \frac{7}{12} - \frac{\frac{3}{4}\ e^{-3t} - \frac{4}{3}\ e^{-4t}}{-1}\right]$✓

(2 × 3) (6)

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[10]

QUESTION 7

7.1 $\frac{C(s)}{R(s)} = \frac{12}{s^2+7s+12}$

$\frac{C(s)}{R(s)} = \frac{12s}{(s+3)(s+4)}$ ✓

NB: $(t) = 2t$

Therefore: $R(s) = \frac{2}{s^2}$

$C(s) = \frac{12s}{(s+3)(s+4)} \times \frac{2}{s^2}$ ✓

$C(s) = \frac{24}{s(s+3)(s+4)}$ ✓

$C(t) = \frac{24}{(3\times4)}\left[1 + \frac{3e^{-4t} - 4e^{-3t}}{4 - 3}\right]$ ✓

$C(t) = \frac{24}{12}\left[1 + \frac{3e^{-4t} - 4e^{-3t}}{1}\right]$

$C(t) = 2\left[1 + 3e^{-4t} - 4e^{-3t}\right]$ ✓

$C(t) = 2 + 6e^{-4t} - 8e^{-3t}$ ✓ (6)

7.2 $\lim_{t\to 0} \{C(t)\} = s \to \alpha\{C(s)\}$
$t = 0$

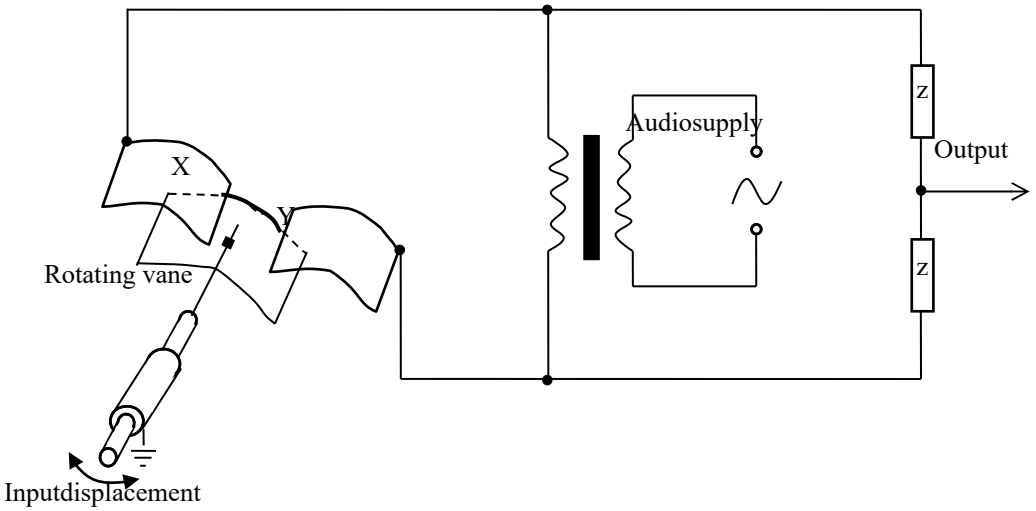
$= 2 + 6e^{-4\times0} - 8e^{-3\times0}$ ✓
$= 2 + 6 - 8$
$= 0$ ✓ (2)

7.3 $\lim_{n\to\infty} \{C(t)\} = s \to 0\{C(s)\}$
$e^{\alpha} = 0$

$= 2 + 6(0) - 8(0)$ ✓
$= 2$ *units* ✓ (2)
[10]

QUESTION 8

8.1



✓✓ (5)

8.2
- As the shaft starts to rotate, the angular displacement of the shaft and the rotating vanes is sensed.
- This device is electrically balanced if the capacitance centres (the rotating vane is the centre) of X and Y are equal.
- The angular displacement of the vane increases the capacitance of Y and decreases that of X or vice versa.
- A bridge supplied by the audio source senses the magnitude and direction of any unbalance.
- This output is applied to an error-voltage amplifier to restore this imbalance by changing the direction of the angular displacement. (5)

[10]

QUESTION 9

9.1
- Power transmission
- Lubrication
- Cooling (3)

9.2 A rotary pump continuously scoops liquid from the pump chamber and a centrifugal pump imparts velocity to a stream of liquid. (2)

9.3 It gives a more positive action and is not influenced by any disturbing forces due to its small inertia and greater rigidity. (2)

9.4
- Rotary vane actuator
- Rack-and-pinion actuator (2)

9.5 Derivative control controls the rate of the error in respect to time. (1)

[10]

QUESTION 10

10.1 10.1.1 Straight line
10.1.2 Ellipse
10.1.3 Circle
10.1.4 Ellipse

(4 × 1) (4)

10.2 10.2.1



✓

10.2.2



✓

10.2.3



✓

(3 × 1) (3)

10.3 $$\frac{f_H}{f_V} = \frac{Number\ \ of\ \ vertical\ \ tangencies}{Number\ \ of\ \ horizontal\ \ tangencies}$$

$$f_V = \frac{Number\ \ of\ \ horizontal\ \ tangencies}{Number\ \ of\ \ vertical\ \ tangencies} \times f_H \checkmark$$

$$f_V = \frac{3}{1} \times 1\ \ 000 \checkmark$$

$$f_V = 3\ \ 000\ \ H_z \checkmark$$

(3)

[10]

TOTAL: 100

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